

Micro-macro changepoint inference for periodic data sequences

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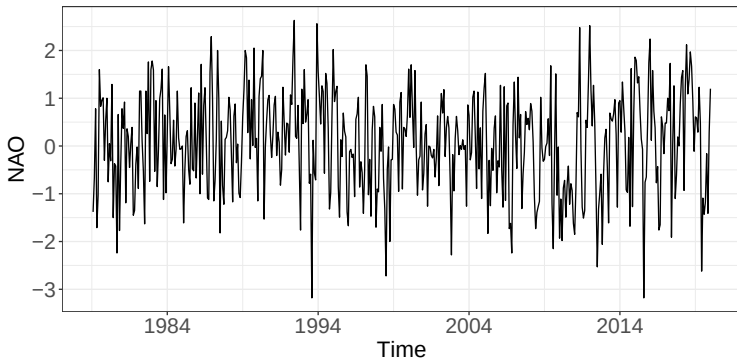
IISA, June 2025

- Motivation
- What are changepoints?
- Micro changepoint model
 - Bayesian Implementation
 - Frequentist Implementation
- Macro changepoint model
- Simulations
- Application

Motivation

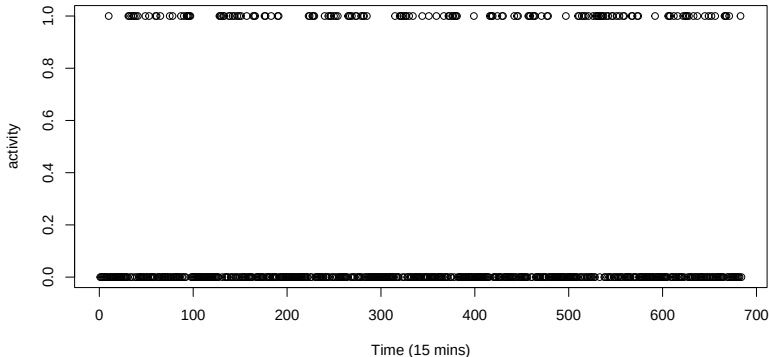
- How can we make the most of highly granular time series?
- How can we study changes in periodic and cyclical behaviours?

NAO fluctuations (1979–2020)



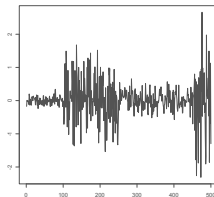
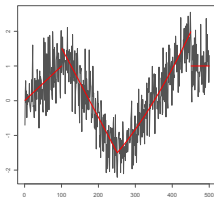
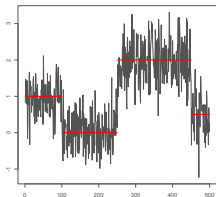


- How can we study changes in periodic and cyclical behaviours?



What are changepoints?

For data $y_1, \dots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \dots, y_\tau$ are different from $y_{\tau+1}, \dots, y_n$ in some way. Traditional changepoints include: mean, regression, variance.

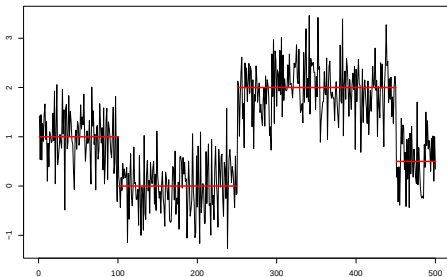


Change in mean

Assume we have time-series data where

$$Y_t | \theta_t \sim N(\theta_t, 1),$$

but where the means, θ_t , are piecewise constant through time.



We want to infer the number and position of the points at which the mean changes. One approach:

Likelihood Ratio Test

To detect a single changepoint we can use the likelihood ratio test statistic:

$$LR = \max_{\tau} \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$

We infer a changepoint if $LR > \beta$ for some (suitably chosen) β . If we infer a changepoint its position is estimated as

$$\tau = \arg \max \{ \ell(y_{1:\tau}) + \ell(y_{\tau+1:n}) - \ell(y_{1:n}) \}.$$

This can test can be repeatedly applied to new segments to find multiple changepoints.

Define m to be the number of changepoints, with positions $\tau = (\tau_0, \tau_1, \dots, \tau_{m+1})$ where $\tau_0 = 0$ and $\tau_{m+1} = n$.

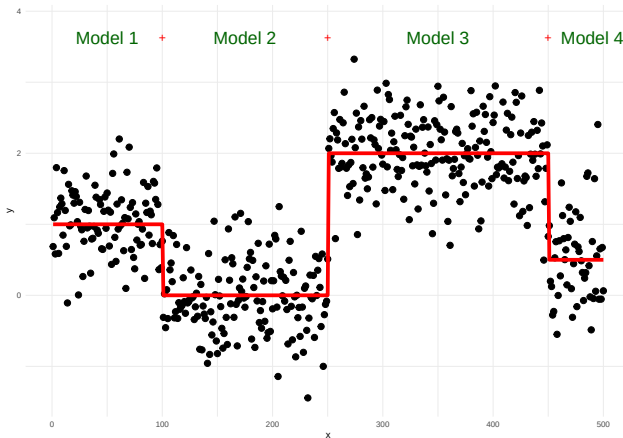
Then one application of the Likelihood ratio test can be viewed as

$$\min_{m \in \{0,1\}, \tau} \left\{ \sum_{i=1}^{m+1} [-\ell(y_{\tau_{i-1}:\tau_i})] + \beta m \right\}$$

Repeated application is thus aiming to minimise

$$\min_{m, \tau} \left\{ \sum_{i=1}^{m+1} [-\ell(y_{\tau_{i-1}:\tau_i})] + \beta m \right\}$$

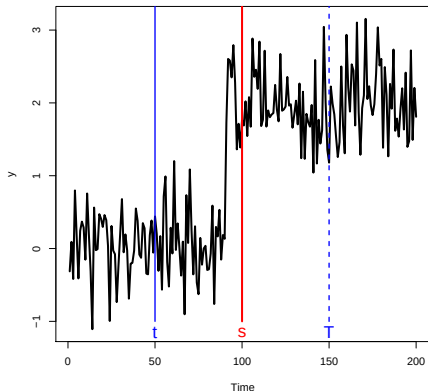
Problem



- How many changes?
- Where are the changes? 2^{n-1} possible solutions!

PELT in a nutshell

- Dynamic programming allows us to only worry about the location of the *last* change.
- Pruning means that as we go through the data we are smart about which locations are potential last change locations.



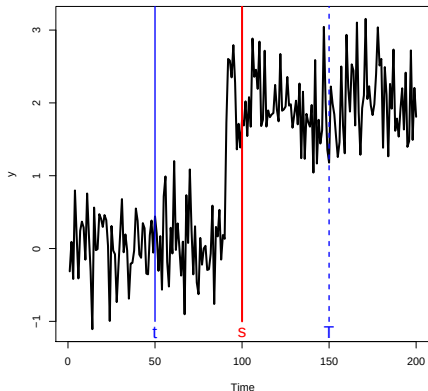


Let $0 < t < s < T$, if

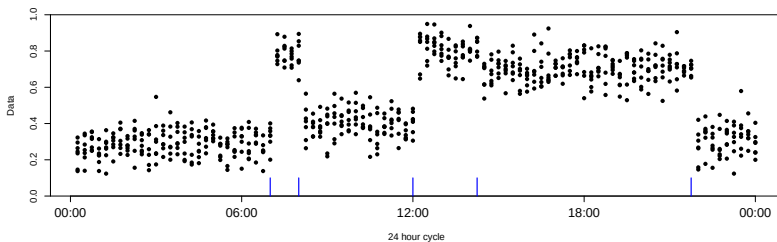
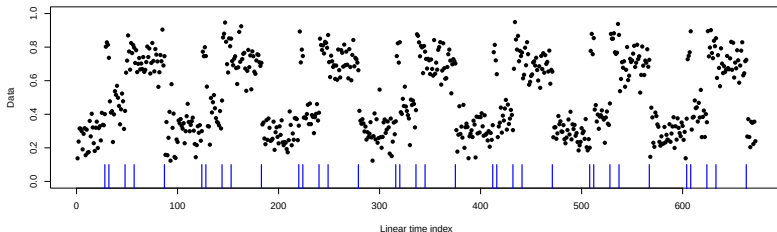
$$F(t) + \mathcal{C}(y_{(t+1):s}) < F(s)$$

then at any future time $T > s$, t
can never be the optimal last
changepoint prior to T .

We can prove that, under certain
regularity conditions, the expected
computational complexity will be
 $\mathcal{O}(n)$.

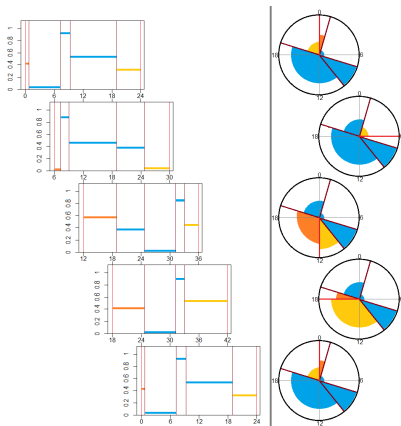


Periodic Changepoints



- Binary segmentation & PELT
 - Assumes conditional independence ...
 - ... but dependence exists across days.
- Multivariate time series
 - Stack days to align changepoints ...
 - ... but what should 'time zero' be?

- Circular changepoint
 - For circular data (e.g. direction)
...
 - ... but time axis is linear.
- Circular Binary Segmentation
 - local time axis is circular ...
 - ... but just joins up ends not each day.
- Cyclic-HMM – Achieves periodicity via a structured transition matrix ... but period duration is stochastic.



Create method for periodic and global changepoint detection

- Wide range of data structures (assume likelihood available)
- Present frequentist method for local periodic changes
- Use PELT to extend to global changes in periodic

Create layered visualizations for ease of inference.

We denote changepoints as micro (periodic) and macro (global).

Periodic level changepoint detection

Challenge: How to incorporate circular nature

Soution: Reframe $t = cN + i$, $c \in \mathbb{N}_0$ and $i \in \{0, 1, \dots, N - 1\}$

Challenge: How to detect using (c, i) instead of t ?

Challenge: There is no single changepoint setting anymore!

- $\{X_t\}_{t \geq 0}$ — Real-valued time series.
- $N \in \mathbb{N}$ — Given period length.
 - $t = cN + i$ where $c \in \mathbb{N}_0$ and $i \in \{0, 1, \dots, N - 1\}$.
- $m \geq 1$ — Number of segments partitioning the period.
- $\tau_1, \dots, \tau_m \in \{0, 1, \dots, N - 1\}$ — Changepoint positions.
 - Ordered: $\tau_{j-1} < \tau_j$.
 - Periodic: $\tau_{m+1} = \tau_1$.
 - Minimum separation: $\tau_j - \tau_{j-1} \geq l$ for $l > 0$.
- μ_j for $j \in \{1, \dots, m\}$ — Segment means.
- σ_j^2 for $j \in \{1, \dots, m\}$ — Segment variances.

$X_t \sim \text{Normal}(\mu(t), \sigma^2(t))$ for $t \geq 0$ where
 $\mu(t) = \sum_{j=1}^m \mu_j \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$ and $\sigma^2(t) = \sum_{j=1}^m \sigma_j^2 \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$.

Frequentist

Challenge: No single changepoint setting to build upon.
Standard:

$$\begin{aligned} c_{0,t}^m &= \min_{\tau_1, \dots, \tau_m} \sum_{j=1}^{m+1} \mathcal{C}(x_{(\tau_{j-1}+1):\tau_j}) \\ &= \min_{\tau_{m-1}} [c_{0,\tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1):t})]. \end{aligned}$$

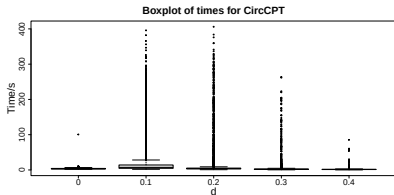
Circular:

$$\begin{aligned} c_{0,t}^2 &= \min_{\tau_1, \tau_2} [\mathcal{C}(x_{(\tau_1+1):\tau_2}) + \mathcal{C}(x_{(\tau_2+1):\tau_1})] \\ c_{0,t}^m &= \min_{\tau_{m-1}} [c_{0,\tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1):t})] \quad m = 3, \dots, M \end{aligned}$$

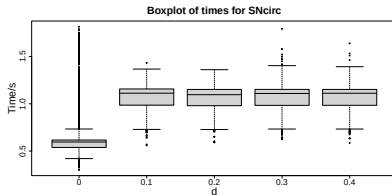
Correct detection ($1-\alpha$ and power rates)

$d \backslash \tau_1$	4	8	16	24	32	48
0.0	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.000	0.004	0.042	0.102	0.123	0.156
0.2	0.177	0.467	0.550	0.540	0.519	0.570
0.3	0.842	0.846	0.828	0.856	0.866	0.873
0.4	0.973	0.975	0.971	0.972	0.966	0.976

Simulation Results

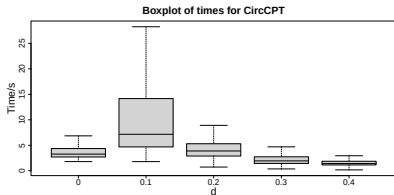


Bayesian

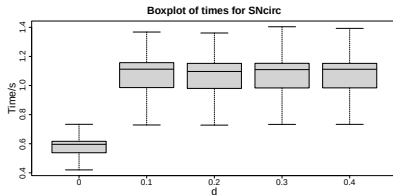


Frequentist

Simulation Results



Bayesian



Frequentist

Macro Level Changepoint Detection

- Treat the micro-level detection as a “fit” for a segment ...
- ... thus we can use traditional, linear time algorithms.

We use PELT which optimizes:

$$\sum_{i=1}^{q+1} [\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i})] + \beta q. \quad (1)$$

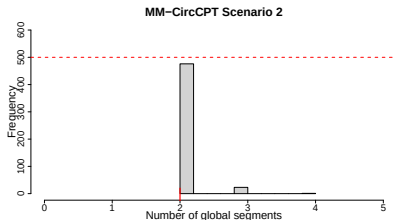
where q is the number of macro level changepoints and $\mathcal{C}(\cdot)$ is the (MAP) likelihood:

$$\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i}) = -2 \sum_{j=1}^{\hat{m}_i} \sum_{t \in B_{i,j}^*} \log f(y_t | \hat{\mu}_{i,j}, \hat{\sigma}_{i,j}^2)$$

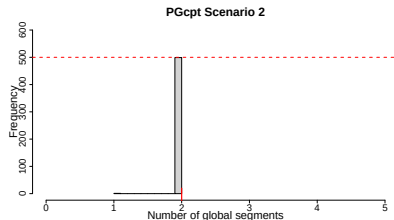
Simulations

See pre-print for full simulation study.

Simulation Results



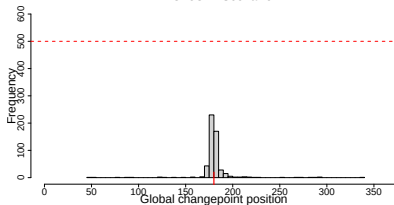
Bayesian



Frequentist

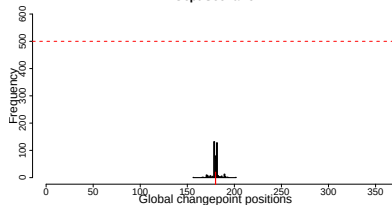
Simulation Results

MM-CircCPT Scenario 2



Bayesian

PGcpt Scenario 2

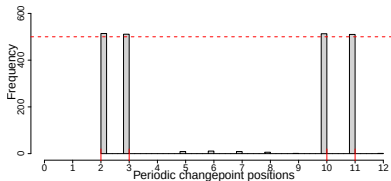


Frequentist

Simulation Results

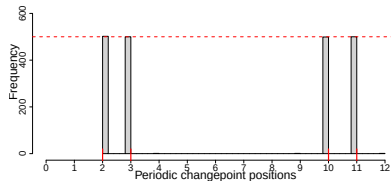


MM-CircCPT Scenario 2



Bayesian

PGcpt Scenario 2



Frequentist

Application



NAO fluctuations (1979–2019)

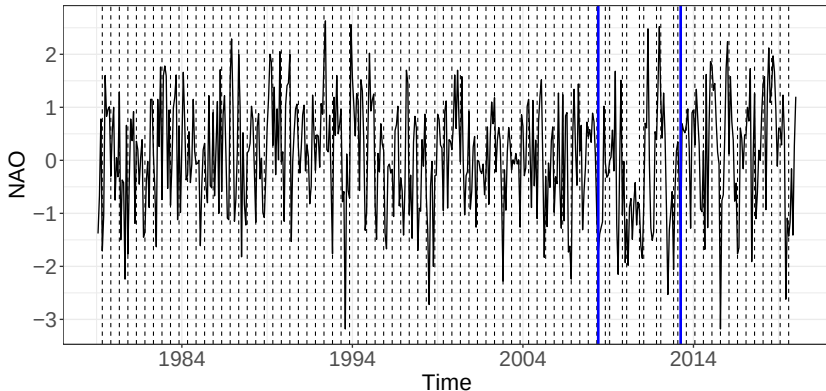
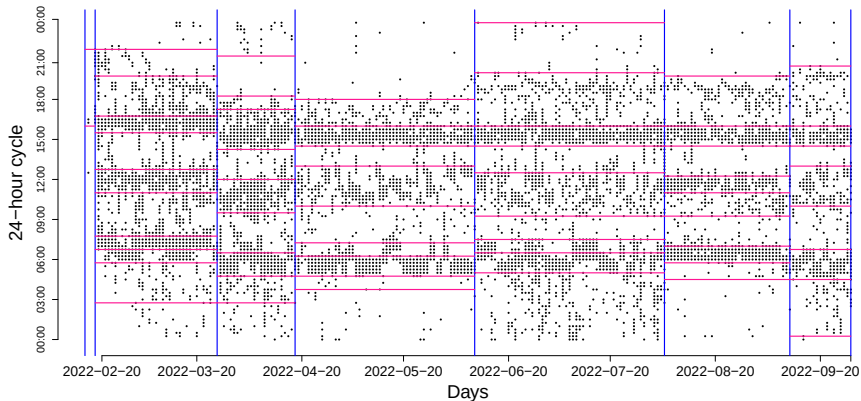


Figure: NAO data and Changepoint positions using $pen = 4 \log(n)$

Macro-level segment	Micro-level changepoints
1979-2008	April and October
2008-2013	May and August
2013-2019	April and October

Table: Macro- and micro-level changepoint positions for the NAO index.

Linear time plot of Howz data

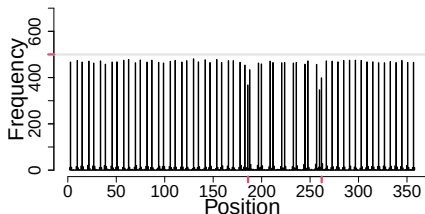


Overview:

- A new approach for identifying within-period changepoint events for time series.
Taylor, S. A. C., Killick, R., Burr, J. and Rogerson, L. (2021)
[‘Changepoint Detection within Periodic Binary Time Series’](#). JRSS:Series C.
- Bayesian pooling of evidence across multiple periods by applying a circular perspective to time.
- Introduced the micro-macro level changepoint approach
Ushakova, A., Taylor, S. A. C., Killick, R. (2023)
[‘Micro-Macro Changepoint Inference for Periodic Data Sequences’](#). JCGS.
- Bayesian and Frequentist viewpoints for a wide set of data types
Li, O., Killick, R. (2025+)
[‘Detecting changes in periodic data’](#). Submitted



Scenario 11



Scenario 7

